SIMULATION REPORT

Simulation experiment of the calibration process

As shown in, the laser vision sensor is driven by the simulated trajectory to measure the simulated straight weld seam. The simulated edge's start point and end point are ${}^{B}P_{start} = (2100.00, 5.00, 900.00)mm$ and ${}^{B}P_{end} = (2100.00, 25.00, 900.00)mm$, respectively. The simulation's parameters are shown in Table I, and ${}^{B}T_{E}^{t-t_d}$ is generated according to an S-shaped trajectory. To simplify the LVS's model, the light plane is set as the X-O-Y plane of the camera coordinate system. The intersection of the light plane and the edge is the measurements ${}^{C}P_{f}$ of the LVS in robotic hand-eye system. The robot scans the fillet weld along an S-shaped trajectory defined in, under 9 different attitudes. In addition, the initial values of the nonlinear optimization are shown in Table I.

$$\begin{cases} X(t) = v_x \cdot t + X_0 \\ Y(t) = H \cdot \sin(t \cdot 1.5\pi) + Y_0 \\ Z(t) = 0.5H \cdot \sin(t \cdot 1.5\pi) + Z_0 \end{cases} \tag{1}$$

where (X_0, Y_0, Z_0) is the starting point of the calibration path and needs to be determined through robot teaching. H is set to 20mm, and v_x is set to 20.0 mm/s.

Through the calibration results of the simulation, as illustrated in Fig. 2, all parameters converge to the true values, demonstrating the effectiveness of the proposed method.

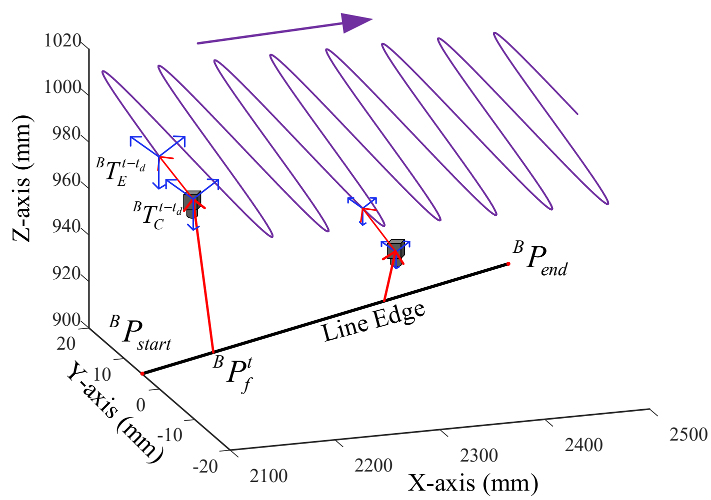

Fig. 1. The simulation schemes.

TABLE I

THE TRUE VALUES AND INITIAL VALUES FOR THE NONLINEAR OPTIMIZATION

Parameters	True values	Initial values
Hand-Eye Rotation	${}^{E}_{C}R = \mathrm{I}_3$	${}^{E}_{C}R^{0} = \mathrm{Exp}(0.01, 0.02, 0.02)$
Hand-eye Translation	${}^{E}P_{C} = [10, 20, 30]^{\mathrm{T}}\ mm$	${}^{E}P_{C}^{0} = [11, 22, 33]^{\mathrm{T}}\ mm$
Plucker coordinates	${}^{B}\omega = (0.08024, 0.02410, 0.03676)$, ${}^{B}m = 851.213$	${}^{B}\omega^{0} = (0.0805, 0.02356, 0.03596)$, ${}^{B}m^{0} = 850.016$
The camera time-offset	$t_d = 0.012s$	$t_d^0 = 0.005s$

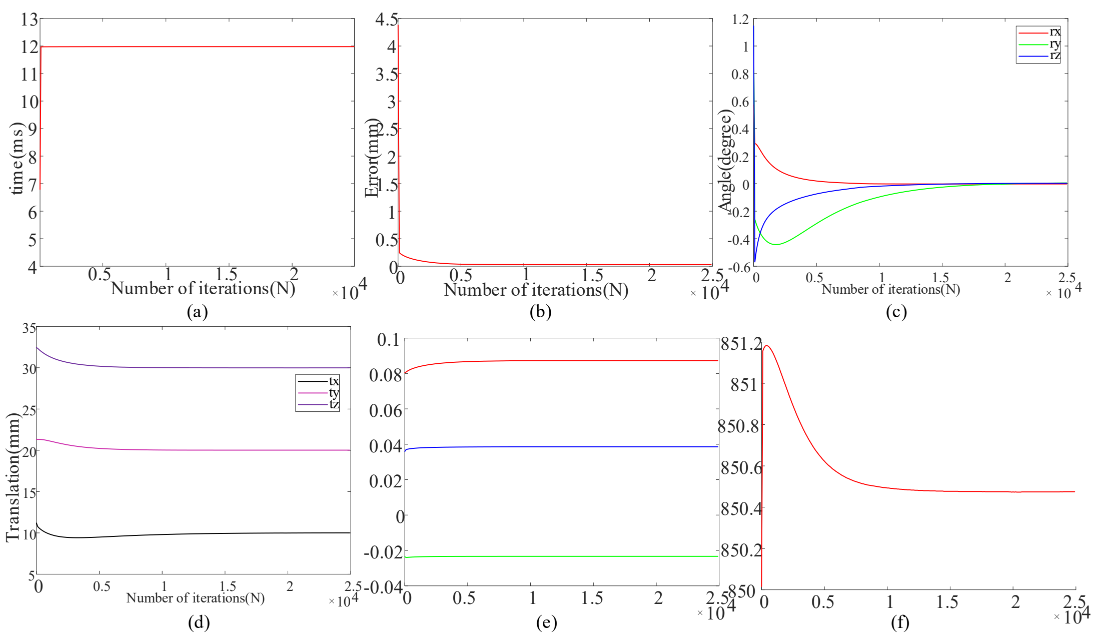

Fig 2. Values of the estimated parameters during the optimization. (a) Values of the camera time-offset. (b) Values of the RMSE. (c) Values of the hand-eye rotation. (d) Values of the hand-eye translation. (e) and (f) Value of the Plucker coordinates of the line.